

Powers and Exponents

Flagship

Lesson 6-1-flagship

Name: _____

Date: _____

Class: _____

SOUND STUDIO MISSION

| The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$5^3$$

$$5 \times 5 \times 5$$

In 2^3 , the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

Write the definition:



base
bottom side

In 5^2 , the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

Write the definition:

$$5^3$$

$$5 \times 5 \times 5$$

$10^3 = 10 \times 10 \times 10 = 1,000$ — read as '10 to the third power' or '10 cubed'

Power

Write the definition:

$$3x + 5$$

no equal sign

Evaluate 3^4 : write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

Evaluate

Write the definition:

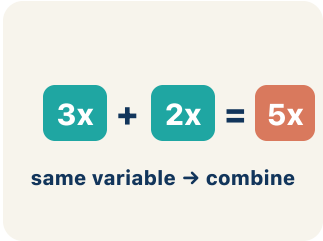

$$4x$$

coefficient = 4

In $3x$, the 3 is the coefficient — if $x = 4$, then $3x = 3 \times 4 = 12$

Coefficient

Write the definition:


$$3x + 2x = 5x$$

same variable $\rightarrow$ combine

$4x$ and $2x$ are like terms (both have x); $4x$ and $4x^2$ are NOT like terms (different powers)

Like terms

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write and evaluate numbers in exponent form using a base and a power.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Exponent

Base

Power

Evaluate

Coefficient

Like terms

1 The small raised number that shows how many times the base is multiplied is the

_____.

2 The number being multiplied repeatedly in a power, like the 3 in 3^4 , is the

_____.

3 A number written with a base and an exponent, like 3^4 , is a _____.

4 To find the value of an expression, like $2^3 = 8$, is to _____ it.

5 A number multiplied by a variable, like the 5 in $5x$, is a _____.

6 Terms with the same variable raised to the same power, like $3x$ and $7x$, are

_____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the value of 4^3 ?

- A. 64
- B. 12
- C. 43
- D. 7

Step 1 $4^3 = 4 \times 4 \times 4 = 64.$

 **Answer:** A. 64

 We do — together

Solve this one with your class using the same steps.

Problem: What is the value of 6^2 ?

- A. 36
- B. 12
- C. 62
- D. 8

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which expression shows 5^3 as repeated multiplication?

- A. $5 \times 5 \times 5$
- B. 5×3
- C. $5 + 5 + 5$
- D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Track 2 needs the value of 5^2 . Which number opens the door?

- A. 25
- B. 10
- C. 52
- D. 7

Show your work:

2. The vault display reads 10^4 . In the power 10^4 , which number is the base?

- A. 10
- B. 4
- C. 40
- D. 14

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because $2^6 =$ _____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because _____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4 ?

- A. 81
- B. 12
- C. 34
- D. 64

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A. $36 - 6^2 = 6 \times 6 = 36$.
8. **You Try (notes):** A. $5 \times 5 \times 5 - 5^3$ means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.
9. **Try It 1:** A. $25 - 5^2$ means $5 \times 5 = 25$, so 25 unlocks Track 2.
10. **Try It 2:** A. 10 — The base is the number being multiplied repeatedly. In 10^4 the base is 10 and the exponent is 4, so $10^4 = 10 \times 10 \times 10 \times 10 = 10000$.
11. **Exit Ticket:** A. $81 - 3^4 = 3 \times 3 \times 3 \times 3 = 81$.